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IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2017.Doi Number

A GAN-based Multi-Sensor Data Augmentation Technique for CNC Machine Tool Wear Prediction

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This research is funded by AIR@InnoHK research cluster of the Innovation and Technology Commission (ITC) of the HKSAR Government.

ABSTRACT The prediction of CNC machine tool wear is limited due to the scarcity of data in industry. A sufficient amount of experimental and real-live factory data is missing to train accurate supervised machine learning prediction models. Considering that collecting a large amount of data in real industrial environments can be a great challenge, this paper investigates the applicability of data augmentation techniques for generating synthetic data. Our paper focuses on a multi-sensor approach for the classification of tool wear states in the milling process. Multi-sensor data fusion is performed in the frequency domain. Extracted features are then used for tool wear classification. In order to increase the amount of training data, a generative adversarial network (GAN) is designed for data augmentation purposes. An early stopping strategy is designed to improve the effectiveness of the proposed GAN. The experimental results show that the proposed GAN helps to significantly improve the performance of tool wear classification models. With the inclusion of GAN, the number of required real-environment industrial data can be reduced. The research shows promising results to supplement experimental data by GAN-based synthetic data for predicting tool wear.

INDEX TERMS Tool wear classification, multi-sensor fusion, generative adversarial network, data augmentation, CNC machines, industrial small data

I. INTRODUCTION

The condition of the cutting tools affects the quality of the product in CNC milling processes. The wear of cutting tools leads to quality deterioration and unexpected time-consuming reworks [1]. Therefore, monitoring the tool wear during milling is crucial to ensure high-quality products. However, in real industrial environments, the tool wear is difficult to be measured directly. On the one hand, direct measurement of tool wear requires a high-quality camera system. On the other hand, in the case of a continuous milling process, direct measurement of tool wear leads to unexpected machine downtime [1, 2]. Thus, indirect tool measurement via external sensors is important, which aims at establishing the relationship between the tool wear and the sensor signals, and meanwhile avoiding interfering the milling process. Nowadays, sensor data-based intelligent monitoring is becoming increasingly popular, as it enables the real-time detection of faults and the application of recent deep learning algorithms, which can help prevent severe systematic failure in advance [3, 4].

In related works, different types of sensors have been explored. Popular sensors include dynamometers (measuring cutting forces), current sensors, accelerometers (measuring vibrations), and acoustic emission (AE) sensors. Milling leads to the degradation of the sharpness of the cutting tool. The blunter the cutting tool, the larger the cutting force, and thus the larger the spindle motor current, which is the energy source of the cutting force [1]. In the milling process, the contact between the cutting tool and the workpiece will produce vibration signals, and the deformation of the workpiece will produce ultrasonic signals that can be detected by AE sensors [2].

On using the dynamometer or the current sensor, the root mean square (RMS) value is shown to be the most effective statistical feature [5, 6]. Different types of neural networks can be used for tool wear prediction, including the multi-

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layer perceptron (MLP), the extreme learning machine, the recurrent neural network (RNN), the long short-term memory (LSTM) RNN, the gated recurrent unit (GRU) RNN, and the convolutional neural network (CNN) [5, 7-9, 10]. Other popular tool wear classification models include the support vector machine (SVM) and the random forest (RF) [11, 12]. On using the accelerometer or the AE sensor, features can be extracted by applying short-time Fourier transform (STFT) [13], wavelet packet transform (WPT) [14] or 1D CNN [15]. Various models can be used to predict tool wear states, including MLP, SVM, and the k nearest neighbor (KNN) model [16, 17]. To predict the raw tool wear values instead of the tool wear states, LSTM RNN, the support vector regression (SVR) model, and the Gaussian process regression (GPR) model can be used [18].

Besides of the single-sensor approaches, combining multiple sensor data for tool wear prediction may also bring benefits. Extracting the time-domain and frequency-domain features from multiple sensors are popular choices [19-22]. Important feature selection can be achieved by Pearson correlation analysis [22, 23]. To predict tool wear, CNN and RNN can be used [19, 20]. Besides, raw time-domain signals from multiple sensors can also be directly processed by CNN for end-to-end tool wear prediction [24].

A tool wear prediction system generally includes three crucial components, i.e., data, features, and models. Although there are plenty of related works on studying features and models, as far as we know, there are few studies focusing on the data component. In real industrial environments, usable data are usually insufficient, which can be a great challenge for data-driven tool wear prediction systems. There can be two reasons leading to data scarcity. First, the tool wear is difficult to measure. Second, it is risky to wait until the worn-out of the tool (the cutting tool is usually replaced in its very early lifetime based on experience, in order to prevent unexpected tool quality reduction due to the absence of an intelligent tool life monitoring mechanism).

Facing the issue of data insufficiency, several strategies can be used to mitigate its effects, including 1) fusing data-driven models with physical laws, 2) utilizing transfer learning for pre-training, and 3) generating new data from existing data [25].

The first strategy is to include the physical knowledge about the milling process into the tool wear prediction models. This requires prior knowledge about the physical milling process. In [26], the Sipos tool wear rate model is fused with the adaptive neuro-fuzzy inference system. The fusion is achieved by a regularized particle filtering algorithm. In [21], a physical-guided model is proposed, which combines a power-function physical model and a GRU RNN. In [27], a hybrid tool wear prediction model is proposed, combining the prediction of an RF regression model into a simulation-based tool wear rate model. In [28], a GPR model based on cutting force features is proposed for tool wear prediction, where the cutting force is estimated using the analytical cutting force model. The GPR model is trained using cutting forces estimated from the analytical model in simulation, and is then tested using cutting forces measured in experiments.

The second strategy is to utilize transfer learning to pre-train the tool wear prediction models using a label-rich dataset. Transfer learning aims at learning knowledge from the source domain (e.g., a dataset with abundant data and labels) and then applying the knowledge to the target domain (e.g., a dataset with few or no labels). In [29], the authors propose a transfer learning method based on optimal transport for tool wear prediction under unmatched working conditions. The method jointly optimizes the source-to-target domain transformation and the target-domain tool wear prediction. In [30, 31], a domain classifier is included in the training of tool wear prediction models to make the features extracted from the source and target domains indistinguishable, to deal with different cutting conditions. In [32], a meta-learning algorithm is used to improve the adaptability of the tool wear prediction model under different working conditions. A neural network-based tool wear prediction model is trained to extract common features under different working conditions and make fast adaptation when the working condition changes. In [33], a neural network model is proposed to predict cutting forces in the milling process based on the cutting parameters. The model is pre-trained using simulated force data and fine-tuned using experimental force data.

The third strategy is to apply data augmentation techniques to enlarge the existing dataset. On using the first strategy, expert knowledge on the physical milling process needs to be known, and the physical model under a specific condition may not be generalizable to other conditions. On using the second strategy, an extra dataset is needed to pre-train the model, which can be difficult to realize in real industrial environments. Therefore, this paper focuses on the third strategy, which aims to generate new data from existing data.

Sensor data are essentially time series data. Data augmentation techniques for time series can be roughly divided into two categories, namely the conventional approaches and the learning-based approaches [34]. As reviewed in [34], for conventional approaches, synthetic data can be obtained by cropping the time-domain signals, injecting noises in the time domain or frequency domain, or masking the frequency spectrum; for learning-based approaches, synthetic data can be obtained by decomposition algorithms, statistical models, or generative models, such as the generative adversarial network (GAN). This paper considers using GAN for data augmentation, which has found various applications and proved to be beneficial [35]. For conventional approaches, the injected noises are essentially unknown; for statistical model-based approaches, prior information about the distribution of data needs to be known. Therefore, GAN is considered as a convenient

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candidate for data augmentation, who has the potential to deal with unknown distributions.

GAN consists of a generator and a discriminator competing with each other. The generator is trained to generate synthetic data, while the discriminator is trained to determine whether the generated data are real or synthetic. For time series data, there exist different choices for the generator and the discriminator. [36] adopts LSTM RNN for the generator and bidirectional LSTM RNN for the discriminator to generate synthetic musical tones. [37] also adopts LSTM RNN for the generator and the discriminator. In addition, it can accept additional inputs, such as data labels, to generate class-specific data. In [38], both the generator and the discriminator adopt the temporal convolutional network (TCN) as the building block, which is a causal 1D CNN able to model long-range dependencies in time series data. In [39], the proposed GAN adopts 1D CNN as the building block for the generator and the discriminator, to synthesize vibration signals for induction motor fault classification. In [40], data generation is performed on the frequency spectrum of the vibration signals. MLP is then adopted for both the generator and the discriminator to generate synthesized spectrums for bearing fault classification.

GAN has found several applications in tool wear prediction. In [41], the researchers design a GAN based on 1D CNN. The input is 3-axis cutting force signals with a length of 600 data points. The finite element method (FEM) is used to simulate the milling process and synthesize more force signals corresponding to different tool wear categories. The proposed GAN is then trained using both experimentally measured signals and simulated signals to generate more time-domain force signals. To perform tool wear classification, time-domain, frequency-domain, and time-frequency domain statistical features are extracted from the force signals, and SVM, RF, and the decision tree (DT) are used for tool wear classification. The experimental results demonstrate the effectiveness of GAN in data generation. It is shown that GAN also outperforms the synthetic minority oversampling technology (SMOTE) as the data augmentation technique. In [42], the researchers apply GAN to solve data imbalance issues. The signal lengths of different sensors are interpolated to be equal, and a CNN-based GAN is used to synthesize more time-domain signals to balance the amount of data for three tool wear categories. In [43], the researchers apply SinGAN [44] to the scalogram of the AE signals and the vibration signals to generate more scalograms to be used for tool wear prediction. Then, image quality parameter (IQP) features are extracted from the scalograms, and fed to LSTM RNN for tool wear prediction.

Different from existing works, in this paper, we propose a multi-sensor data augmentation approach, which utilizes GAN to synthesize more feature vectors instead of synthesizing more time-domain sensor signals. The proposed approach supports multiple sensor types, which provides higher flexibility and potentially improved performance and robustness as compared to single-sensor approaches. The benefits of performing data augmentation in the feature domain instead of the original time domain lie in three aspects. First, since the dimensionality of the extracted features is significantly reduced as compared to the raw time-domain signal, the structure of GAN can be simple, and the training of GAN can be more efficient. Second, the extracted features have a pre-defined fixed dimensionality, while the raw time-domain signals may have different lengths. Third, in a heterogeneous sensor setup, the fusion of extracted features is simpler than the fusion of time-domain sensor signals.

We also propose a practical early stopping strategy to effectively train a GAN. Early stopping is a popular strategy for preventing neural networks from overfitting the training data. Commonly it is used in the training stage of a classification model. For example, [45] performs faulty line detection using telemetry data, where GAN is used for data augmentation. Only the fault detection model is assisted by early stopping, which is achieved by observing the loss on a validation dataset. In several studies, early stopping is also used for training GAN. In [46], early stopping is achieved by evaluating whether the binary classification model trained with GAN-synthetic data has obtained its highest true positive rate. In [47], early stopping is achieved by observing whether the training of GAN has reached equilibrium, where the discriminator loss ceases changing.

Different from most studies, we apply early stopping to the training of GAN instead of the training of the classifier. Different from existing early stopping strategies for GAN, in our approach, early stopping is achieved by observing whether the discriminator loss and the generator loss have reached their local optimum instead of equilibrium. Stopping before equilibrium may bring some benefits as shown in our experiments. In addition, the proposed early stopping strategy does not rely on additional validation datasets or specific classification tasks, and therefore can be more general.

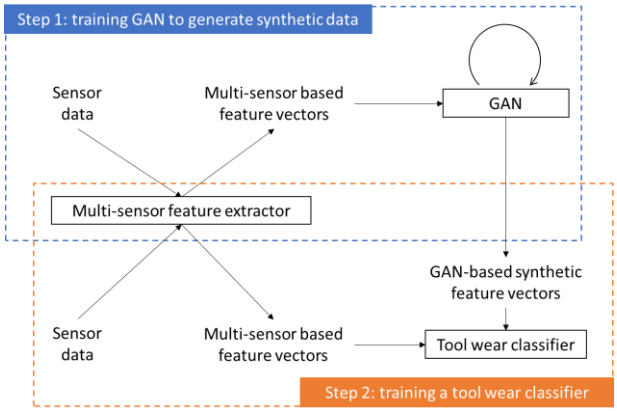

FIGURE 1. Overview of training a GAN-assisted tool wear classification model.

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In this paper, we adopt a multi-sensor approach to classify the tool wear into different states. Specifically, three types of sensors are considered, namely the dynamometer, the accelerometer, and the AE sensor. Frequency-domain features are extracted and combined from different sensor channels. Various sensor combinations and models are investigated in the experiments.

Our major contributions are briefly summarized as follows.

1) We develop a multi-sensor approach for tool wear classification, and design a simple data pipeline combining heterogeneous sensor data in the frequency domain. The paper outlines the experimental results of different sensor combinations and models.
2) We propose a GAN structure and a practical early stopping strategy to effectively train the GAN, so that the synthetic data generated from the trained GAN can help boost the classification accuracy significantly. Experiments are conducted using different classification models. Some limitations of GAN are also analyzed.

The rest of the paper is organized as follows. In section II, the dataset, the feature extraction methods, the tool wear classification models, and the proposed GAN are explained. In section III, tool wear classification experiments are conducted using a publicly available dataset. Single-sensor features and multi-sensor features are investigated by employing different classification models. The effectiveness of GAN and the limitation of GAN are analyzed. A conclusion is drawn in section IV.

II. METHODS

Our proposed data augmentation method generates synthetic feature vectors as a supplement to the existing feature vectors. The synthetic feature vectors together with the existing feature vectors are used to train a GAN-assisted classifier. Interestingly, the training of GAN only requires the extracted feature vectors, which indicates our proposed method will be compatible with various feature extraction methods and classification models. Training a GAN-assisted

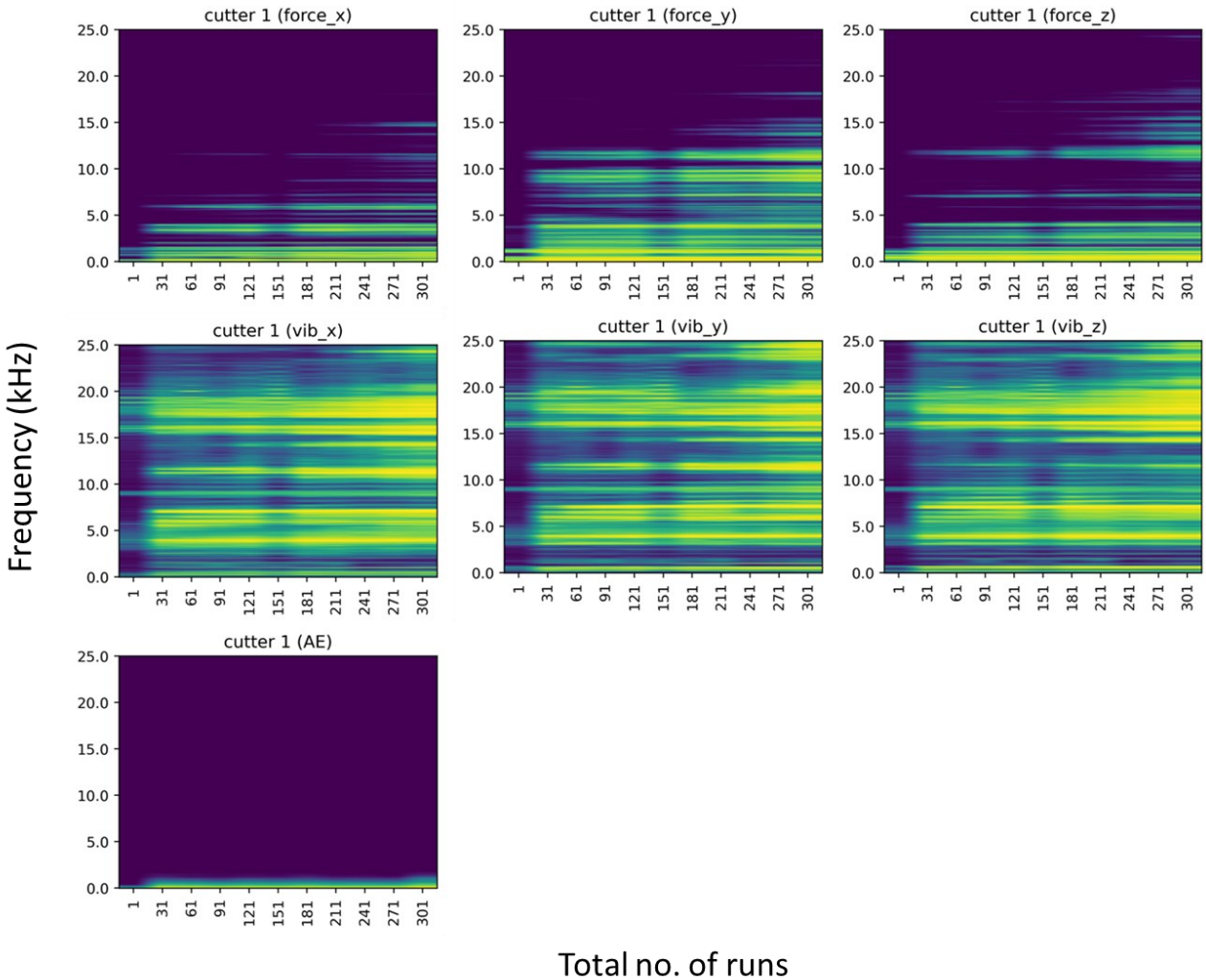

FIGURE 2. **Spectrum changing trend (histogram equalized) of different sensor channels for cutter 1.**

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classifier includes two steps, as shown in Fig. 1. The first step is to train the GAN using existing feature vectors. The second step is to train the GAN-assisted classifier using the existing feature vectors and the synthetic feature vectors.

A. DATA PREPARATION

In this paper, tool wear classification is performed on a publicly available dataset PHM2010 [48]. This dataset consists of the tool wear measurements for 6 cutting tools during the face milling process. The milling process has a spindle speed of 10400 RPM and a feed rate of 1555 mm/min. The radial and axial depth of cut is 0.125 mm and 0.2 mm respectively. More details on the cutting parameters can be found in [48].

Each cutter is used to mill the workpiece for 315 runs (each run lasts for about 4 s). During milling, 3 types of sensors with totally 7 channels are used to record relevant data for the analysis of tool condition. The sensor channels include the x, y, and z directions of the force sensor (denoted as force_x, force_y, and force_z, respectively), the x, y, and z directions of the vibration sensor (denoted as vib_x, vib_y, and vib_z, respectively), and the acoustic emission sensor (denoted as AE). The sampling rate of the data acquisition device is 50 kHz per channel. After each run, the tool flank wear is measured for all the three flutes on the cutter.

In this paper, the data from cutters 1, 4, and 6 are used, whose tool flank wear values are available. The max tool flank wear of the three flutes is used as the target tool flank wear after each run. Since the focus is on the classification of tool wear, the tool flank wear values are further divided into 5 states covering the range of 30 μm to 240 μm, with each class spanning an equal range of 42 μm.

The sensor data and tool wear measurements from cutters 1 and 6 form the training set, as these cutters' tool flank wear values cover the whole range. The data from cutter 4 are used for testing. This split results in a training set of 630 sensor signal segments per channel, and a testing set of 315 sensor signal segments per channel. Each signal segment corresponds to a high-frequency time series.

B. FEATURES

There exist different feature extraction methods for time series data, such as the time-domain statistical features and the time-frequency features. In this paper, considering that the sensor data of one run are of high frequency (50 kHz) and acquired within a short time period (about 4 s), it is suitable to extract the frequency-domain features. This is also supported by observations on the frequency spectrum of the sensor data from different channels. As shown in Fig. 2, there are obvious changes in the frequency spectrum with the total number of runs increasing (i.e., with the total milling time increasing).

Considering that the frequency distribution will not change too much within one run, instead of computing the

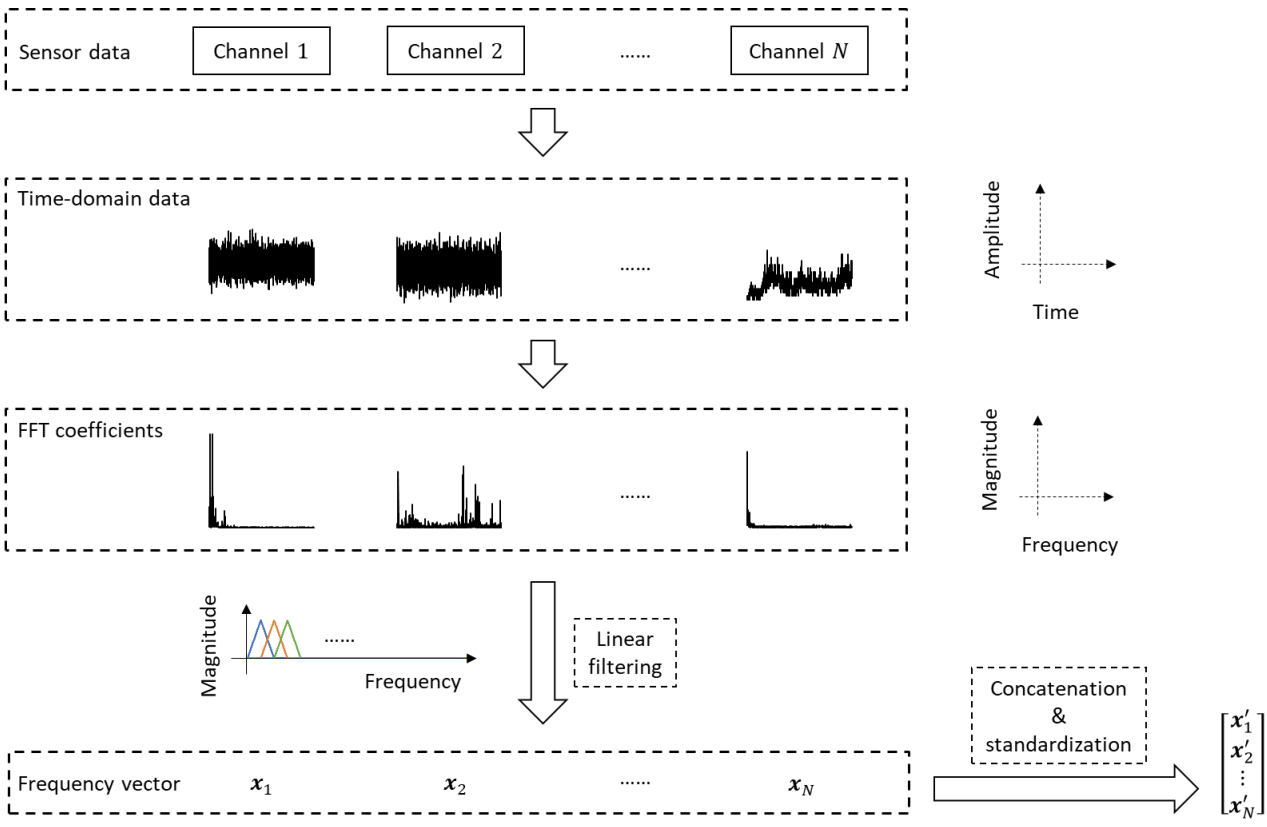

FIGURE 3. **Multi-sensor feature extraction method.**

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spectrogram, we directly compute the fast Fourier transform (FFT) of the signal within one run, and then perform linear filtering using 16 filter bins to further compress the FFT coefficients to have 16 frequency components characterizing the complete frequency spectrum. For vib_x, vib_y and vib_z, all 16 features are kept, as the frequency components span the whole spectrum. For force_x, force_y and force_z, only the first 8 features (corresponding to the low-frequency bins) are kept, as the energy is negligible in high-frequency parts, as seen from Fig. 2. For AE, only the first 4 features are kept, as most energy concentrates in low-frequency parts.

The benefits of computing FFT consist in two aspects. First, the signal energy in the time domain is preserved in the frequency domain. Second, more information is preserved in the raw FFT coefficients than the statistical features of the FFT coefficients (e.g., mean, variance, etc.). The benefits of linear filtering the raw FFT coefficients also lie in two aspects. First, the dimensionality of the filtered frequency vector is only dependent on the number of filters instead of the number of FFT coefficients, which achieves a mapping from a variable-length input to a fixed-length output. Second, linear filtering performs dimensionality reduction, which compresses the data size and improves the computation efficiency.

After computing the feature vector (i.e., the frequency vector) for each sensor channel, the feature vectors of different sensor channels for the same run are concatenated to form a higher-dimensional feature vector. This higher-dimensional feature vector is then standardized to follow a standard Gaussian distribution (with zero mean and unit variance) to prevent extreme values. Both training and testing feature vectors are standardized with respect to the training feature vectors. This feature extraction method is also shown in Fig. 3.

C. CLASSIFICATION MODELS

In this paper, the number of available sensor data and the dimensionality of the feature vector can be considered small. Hence, the following popular classification models are adopted for tool wear classification, including the logistic regression (logit) model, the linear SVM, the MLP, the RF, and the CNN. In the experiments, for MLP, the notation "MLP H*i*-*j*" denotes that the first hidden layer has *i* neurons, and the second hidden layer has *j* neurons. For CNN, the notation "CNN C*i*-*j* H*k*" denotes that the first convolutional layer has *i* channels, the second convolutional layer has *j* channels, and the fully connected layer has *k* neurons. For different structures of MLP, the hidden layers use sigmoid as the activation function, while the output layer uses softmax. For different structures of CNN, each convolutional layer uses a 3×3 kernel, with the rectified linear unit (ReLU) [49] as the activation function, followed by 1×2 max pooling (since the input is a vector, pooling is performed only in one dimension). The fully connected layer uses sigmoid as the activation function, and the output layer uses softmax.

ALGORITHM I
TRAINING OF GAN

Let $X = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \dots \boldsymbol{x}_N\}$ be the original training data, where N denotes the number of training feature vectors.
Let $Z = \{\boldsymbol{z}_1, \boldsymbol{z}_2, \dots \boldsymbol{z}_N\}$ be the input vectors to the generator.
Let $Y = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \dots \boldsymbol{y}_N\}$ be the output synthetic vectors from the generator.
Let $L_{real} = \{[1,0], [1,0], \dots\}$ be the target labels for the original training feature vectors.
Let $L_{syn} = \{[0,1], [0,1], \dots\}$ be the target labels for the output synthetic vectors from the generator.
Let M_g denotes the parameters of the generator.
Let M_d denotes the parameters of the discriminator.

For i = 1 to iteration:
  Randomly generate Z using a standard Gaussian distribution
  Compute synthetic data as $Y = M_g(Z)$
  For j = 1 to epoch:
    Compute the loss between discriminator outputs $M_d(\{X, Y\})$ and target labels $\{L_{real}, L_{syn}\}$, and then adjust the parameters of M_d
  Randomly regenerate Z using a standard Gaussian distribution
  For j = 1 to epoch:
    Compute the loss between discriminator outputs $M_d(M_g(Z))$ and target labels $\{L_{real}\}$, and only adjust the parameters of M_g

D. DATA AUGMENTATION WITH GAN

1) GAN TRAINING PROCEDURES

GAN consists of a generator and a discriminator [50]. The generator aims at synthesizing new data, while the discriminator aims at determining whether the data are real or synthetic. The generator accepts a random vector as input and produces a synthetic feature vector as output. The discriminator accepts a feature vector as input and produces a label telling whether the input feature vector is real or synthetic.

The generator and the discriminator are trained alternately. When the generator is being trained, the parameters of the discriminator are frozen; when the discriminator is being trained, the parameters of the generator are frozen. Suppose the real data have a label of 1 and the synthetic data have a label of 0. Then, the discriminator is trained to predict the real data to have label 1 and the synthetic data to have label 0, while the generator is trained to synthesize new data such that the discriminator will predict the synthetic data to have label 1. This alternate training procedure is repeated for a number of iterations until reaching equilibrium (i.e., both the generator and the discriminator cannot be improved) or satisfying some stopping criteria. The training algorithm of GAN is also shown in Algorithm I.

The training process for GAN and the training process for the classification model with data augmentation are shown in Fig. 4, where $\boldsymbol{x}_n^{(k)}$ denotes the n-th feature vector coming from the real data, $\boldsymbol{z}_n^{(k)}$ denotes the n-th random input vector to the GAN, $\boldsymbol{y}_n^{(k)}$ denotes the n-th synthetic feature vector. The superscript k denotes which class the feature vector

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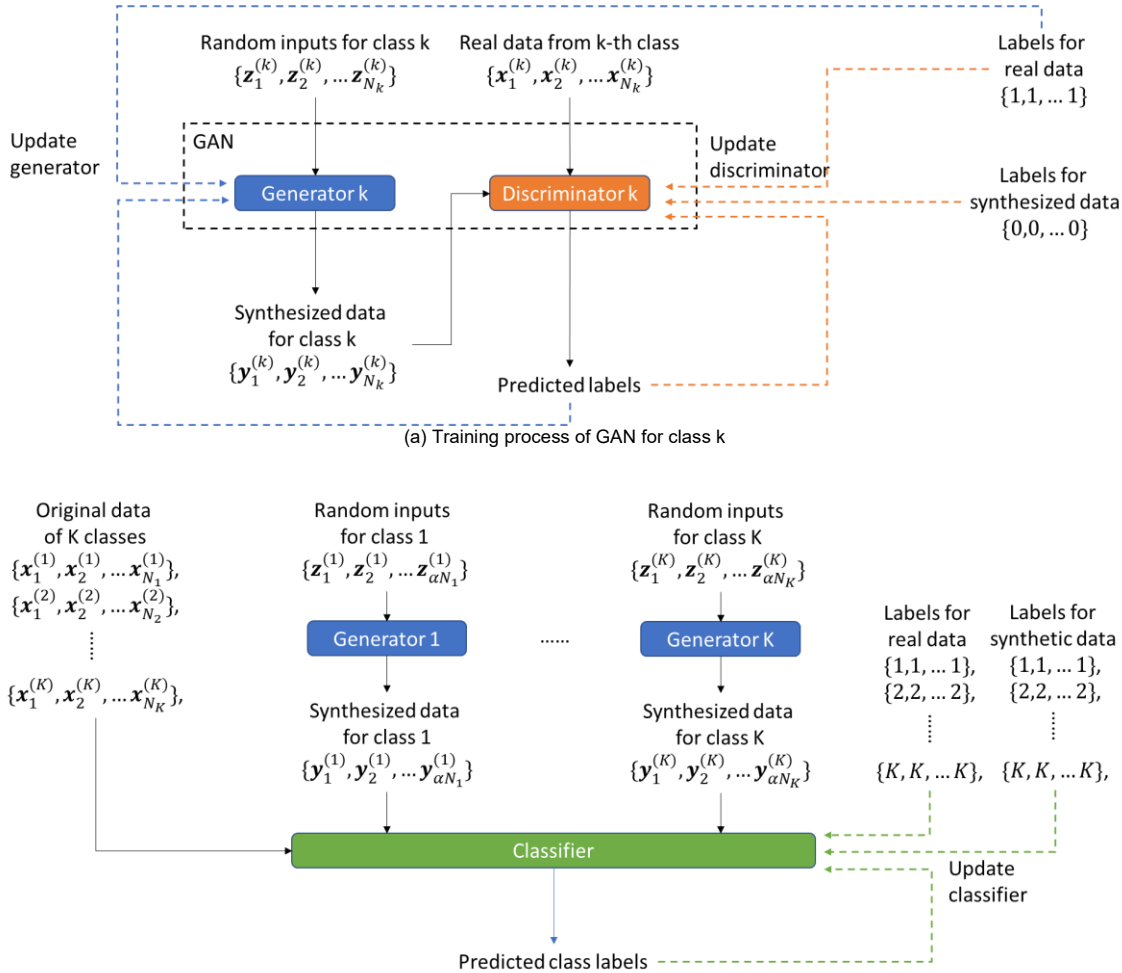

FIGURE 4. Training procedures for GAN and the tool wear classification model.

belongs to (i.e., which tool wear state the feature vector corresponds to). N_k denotes the number of real feature vectors in class k. In Fig. 4(b), α in the subscript denotes the ratio between the number of synthetic data and the number of real data. This ratio can be any non-negative integer, since the number of synthetic feature vectors are unlimited. K is the total number of classes (i.e., total number of tool wear states), and K=5 in this paper.

2) GAN MODEL STRUCTURES

In the proposed GAN, the discriminator is an MLP with a hidden layer of 256 neurons and an output layer of 2 neurons. The hidden layer uses ReLU as the activation, and the output layer uses softmax. The loss function uses the binary cross-entropy loss, and the optimizer is Adam [51]. The generator is also an MLP with a hidden layer of 256 neurons and an output layer of D neurons, where D is the dimensionality of the feature vector. The hidden layer also uses ReLU as the activation. If the feature vector is the frequency vector obtained from a single sensor channel, the output layer will use ReLU as the activation, since the elements in a frequency vector are non-negative. If the feature vector is the concatenated frequency vector obtained from multiple sensor channels, the output layer will not use any activation, since after standardization, the elements in the feature vector will have both positive and negative values.

III. EXPERIMENTS

A. TOOL WEAR CLASSIFICATION WITH DIFFERENT SENSORS AND CLASSIFIERS

In this part, the classification performance using different sensor channels is compared. Different classification models are employed, including the linear SVM, logit, MLP, RF, and CNN. For MLP, different structures are investigated, including "MLP H4", "MLP H4-4", and "MLP H8". For CNN, different structures are investigated, including "CNN C4 H4", "CNN C8 H8", "CNN C4-4 H4", and "CNN C8-8 H8". In fact, since logit uses softmax as the output but does not have any hidden layer, it can be treated as a special MLP with zero hidden layer. On using MLP and CNN, the optimizer is Adam, the loss is the cross-entropy loss, the learning rate is 0.001, and the number of training epochs is 50000. SVM, logit and RF are implemented using scikit-

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TABLE I
CLASSIFICATION ACCURACY OF DIFFERENT SENSOR CHANNELS AND DIFFERENT CLASSIFIERS

	Sensor channels	Feature dimensions	GAN input dimensions	Classifiers				
				SVM	Logit	MLP H4	MLP H4-4	MLP H8
Single sensor	vib_x	16	4	0.2	0.321	0.473	0.473	0.362
	vib_y	16	4	0.375	0.257	0.267	0.444	0.238
	vib_z	16	4	0.133	0.225	0.225	**0.568**	0.248
	force_x	8	4	0.327	0.444	0.276	0.362	0.292
	force_y	8	4	0.254	0.203	0.241	0.254	0.251
	force_z	8	4	0.190	0.311	0.117	0.117	0.121
	AE	4	4	0.124	0.124	0.197	0.197	0.184
Multi sensor	vib_xyz	48	12	**0.498**	0.438	0.222	0.343	0.343
	force_xyz	24	12	0.317	0.346	0.197	**0.378**	0.251
	vib_xyz & force_xyz & AE	76	24	0.378	0.378	0.184	0.362	0.283

	Sensor channels	Feature dimensions	GAN input dimensions	Classifiers				
				RF	CNN C4 H4	CNN C8 H8	CNN C4-4 H4	CNN C8-8 H8
Single sensor	vib_x	16	4	0.413	**0.521**	0.467	0.467	0.365
	vib_y	16	4	0.502	0.298	0.267	**0.629**	0.622
	vib_z	16	4	0.181	0.213	0.222	0.324	0.184
	force_x	8	4	0.346	0.362	0.241	0.454	**0.622**
	force_y	8	4	**0.276**	0.241	0.263	0.168	0.257
	force_z	8	4	**0.371**	0.149	0.127	0.210	0.184
	AE	4	4	0.251	0.232	0.238	0.219	**0.257**
Multi sensor	vib_xyz	48	12	0.422	0.286	0.444	0.435	0.416
	force_xyz	24	12	0.260	0.162	0.222	0.359	0.165
	vib_xyz & force_xyz & AE	76	24	0.232	0.365	**0.381**	0.371	0.362

TABLE II
COMPARISON OF DIFFERENT SENSOR CHANNELS FOR THE SAME SENSOR TYPE

Sensor type	Sensor channel	No. of best-performing classifiers among the 10 classifiers in Table I
Vibration	X	5
	Y	4
	Z	1
Force	X	8
	Y	1
	Z	1

learn [52], while MLP and CNN are implemented using pytorch [53]. SVM uses default parameters, and RF uses 4 estimators with a max depth of 8.

The classification accuracy results are shown in Table I (the highest accuracy among different classifiers for each feature vector is bolded). In general, the classification performance depends on the choice of features and classifiers. Among different combinations of features and classifiers, "CNN C8-8 H8" and RF produce high accuracy. RF outperforms SVM and logit for single-sensor features, as the former is an ensemble model, while SVM and logit outperform RF for multi-sensor features, when more information is carried by the features. In most cases, CNN outperforms MLP, due to the weight sharing mechanism of the former.

By comparing the performance of different single-sensor features, it can be observed that the vibration sensor is more effective than the force sensor and the AE sensor. This observation is more obvious when comparing the performance of vib_xyz and force_xyz under different classifiers, where vib_xyz consistently outperforms force_xyz, except for "MLP H4-4". For each sensor channel, through counting the number of classifiers that achieve a higher classification accuracy than the other channels of the same sensor type, as shown in Table II, it is found that the x channel is the most effective channel for both vibration sensors and force sensors. This may result from the fact that the cutting tool moves in the x direction.

B. TOOL WEAR CLASSIFICATION AIDED BY GAN

In this part, we investigate whether GAN improves the performance of different combinations of features and classifiers by generating synthetic data to be used for training the classifier.

1) TRAINING OF GAN

GAN is trained using Algorithm I, where the number of iterations varies for different feature vectors, but the number of epochs inside each iteration is set to 2.

In the experiments, one GAN is trained for each class. The class-specific GAN is trained using class-specific training data and then used to generate synthetic data for such specific class. Due to the difference of classes in quantity and feature characteristics, different class-specific GANs may require different numbers of training iterations. In most cases, a learning rate of 0.00001 is used, while in a few cases, a learning rate of 0.0001 is used. A smaller learning rate is preferred. However, sometimes a larger learning rate may help the training reach equilibrium, while a smaller learning rate cannot. In this situation, the larger learning rate is used. During the training of GAN, it is found that if a learning rate cannot make the training reach equilibrium, a smaller

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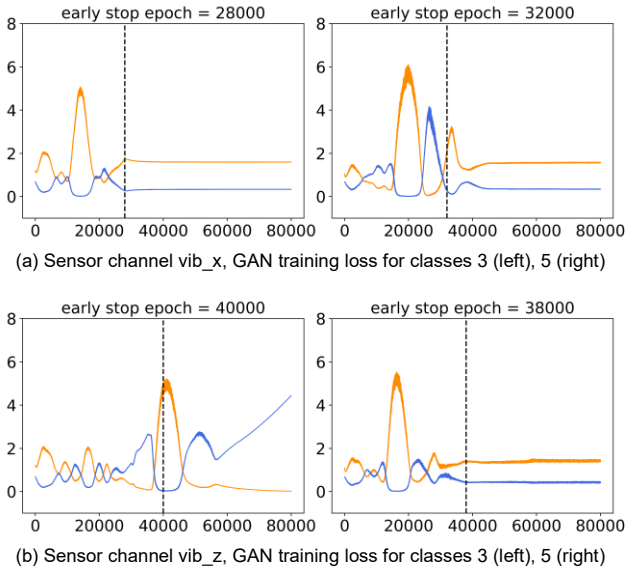

FIGURE 5. Early stopping strategy for training GAN, visualized for different tool wear classes (i.e., tool wear states) on using vibration sensor data. Vertical axis: the binary cross entropy loss per epoch. Horizontal axis: the number of total training epochs. Orange line: discriminator loss. Blue line: generator loss.

learning rate will also not make the training reach equilibrium. Therefore, it is not recommended to use a too small learning rate, which merely slows down the training process.

In the experiments, it is observed that when the training proceeds, the generator loss (i.e., the loss caused by training the generator) and the discriminator loss (i.e., the loss caused by training the discriminator) will fluctuate until equilibrium (i.e., both generator and discriminator losses will not change) or diverge (i.e., the generator loss starts increasing to infinity). In order to prevent the training of GAN from being divergent or simply memorizing the existing data, a simple early stopping strategy is adopted. This early stopping strategy chooses a stopping epoch where the generator loss is neither diverging nor in equilibrium. The stopping epoch can be at a local minimum or the turning point on the training loss curve of the generator. This can be achieved by temporarily saving intermediate GANs in the training process, and choosing an appropriate intermediate version by inspecting the loss curve. An example of the proposed early stopping strategy is shown in Fig. 5. Since the intermediate GANs are saved per 1000 epochs, the stopping epoch is close to but not exactly at the local optimum.

2) GAN ON SINGLE-CHANNEL FEATURES

In this part, the effectiveness of GAN on single-channel features is investigated. For each combination of the sensor channel and the classifier, one GAN is trained for each class. When training the classifier with original training feature vectors and synthetic feature vectors, the ratio between the number of synthetic data and the number of original data varies from 4 to 256, in order to investigate the influence of the quantity of synthetic data. Due to the randomness of the synthesis process, data generation is repeated for 20 times and the best-performing synthetic data are kept. The classification performance with and without GAN is shown in Fig. 6. α is used to represent the data generation ratio, where $\alpha \in [4,16,64,256]$.

For vibration sensor channels as shown in Figs. 6(a) ~ 6(c), when more training feature vectors are generated by GAN, channel y obtains consistent performance improvement under different classifiers, while channel x and channel z cannot obtain performance improvement under RF and logit respectively. This implies vib_y is more sensitive and effective than vib_x and vib_z. For force sensor channels as shown in Figs. 6(d) ~ 6(f), only half of the classifiers can obtain performance improvement when more feature vectors are synthesized by GAN. This implies the force sensor is less effective than the vibration sensor. Among different classifiers, RF and "CNN C4-4 H4" can consistently improve the accuracy with different force sensor channels. For the AE sensor, the performance of MLP and CNN can be significantly improved with the increase of the number of synthetic training feature vectors. A larger number of generated data consistently produces a better performance.

From the experimental results, it is found that the choice of the sensor channel will influence the effectiveness of GAN. Overall, vib_y is the most effective sensor channel with respect to the assistance of GAN. The classification accuracy can be consistently improved to be above 0.6 on using different classifiers, and the highest achievable accuracy is slightly higher than 0.8. Besides, MLP and CNN benefit more from the assistance of GAN than SVM, logit and RF. In addition, a larger number of synthetic data does not always imply a better performance. The consequence of generating more synthetic feature vectors highly depends on the certain combination of the sensor channel and the classifier.

3) GAN ON MULTI-SENSOR FEATURES

In this part, the effectiveness of GAN on multi-sensor features is investigated. For each combination of the feature vector, the classifier, and the desired number of synthetic feature vectors, data generation is repeated for 20 times and the best-performing synthetic data are kept. The classification performance with and without GAN is shown in Fig. 7.

As can be seen from Figs. 7(a) and 7(b), vibration sensor channels are more effective than force sensor channels in terms of benefiting from the synthetic data. With the help of GAN, the highest achievable accuracy of vib_xyz is more than 0.75, while the highest achievable accuracy of force_xyz is about 0.6. Nevertheless, by comparing Fig. 6 and Fig. 7, vib_xyz and force_xyz do not further improve the performance over their single-channel features, such as vib_y and force_y. Besides, combining all sensor types, as shown in Fig. 7(c), does not offer performance improvements over using only the vibration sensor. These findings indicate that combining different sensor channels is not always an optimal choice, since some sensor channels can be more effective

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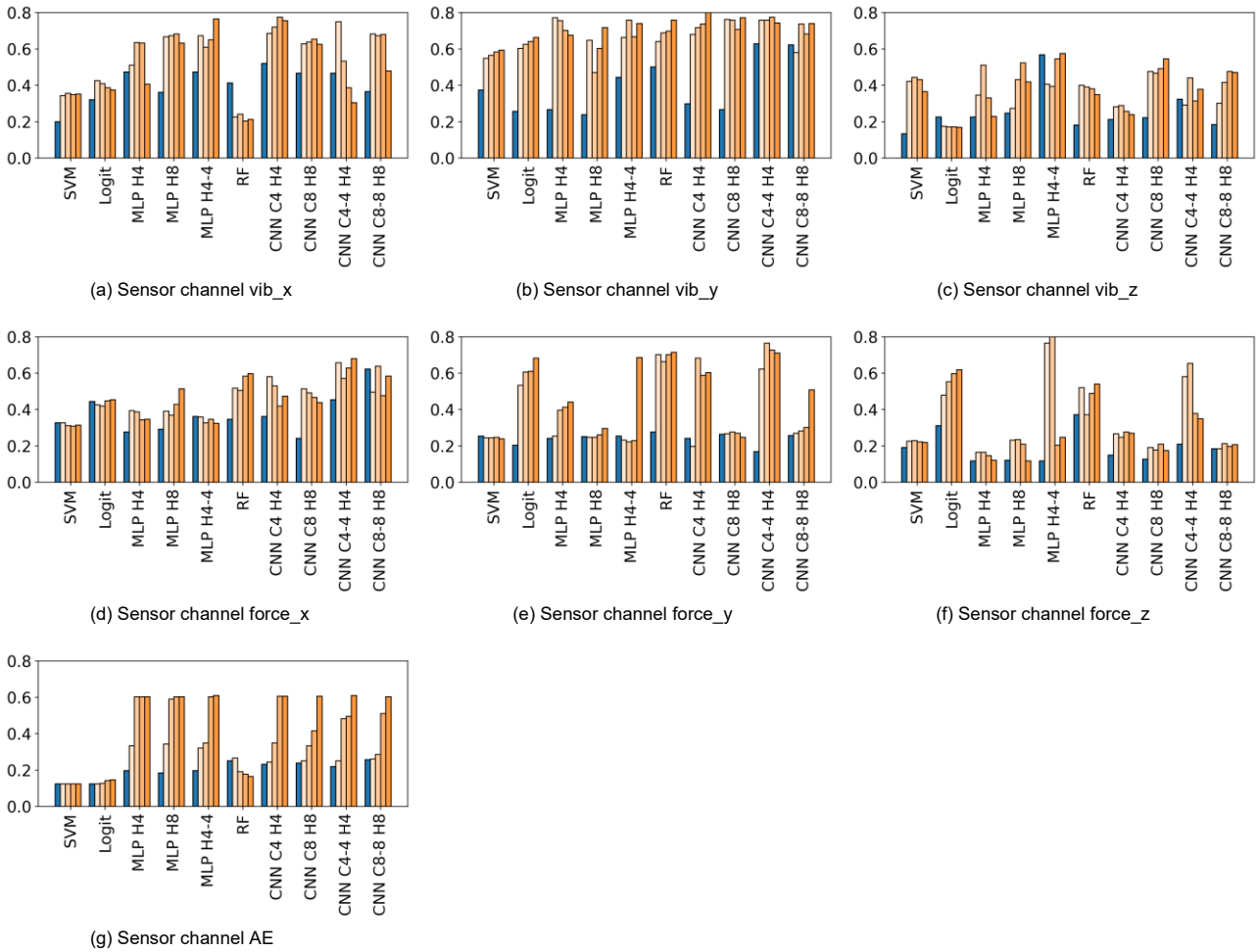


(a) Sensor channel vib_x (b) Sensor channel vib_y (c) Sensor channel vib_z

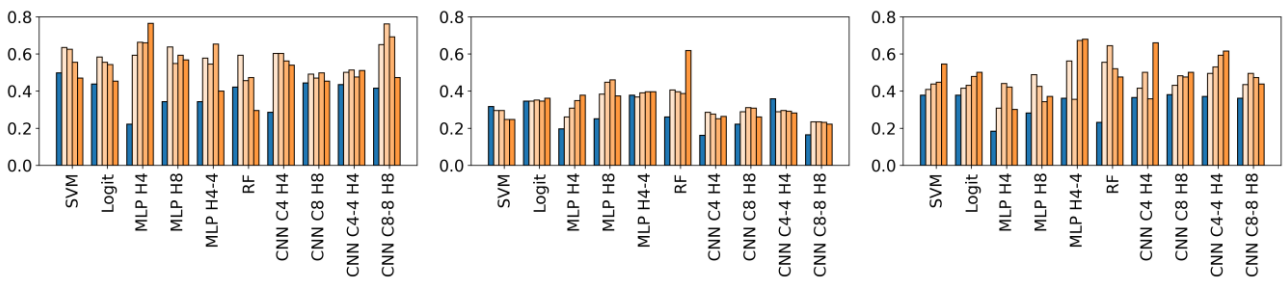
(d) Sensor channel force_x (e) Sensor channel force_y (f) Sensor channel force_z

(g) Sensor channel AE

FIGURE 6. Single-channel approach with and without GAN-based synthetic training data. Horizontal axis: different classifiers. Vertical axis: classification accuracy. Blue bar: classification accuracy without GAN. Orange bars: classification accuracy aided by GAN. Color darkness: colors varying from lighter to darker represent α varying from 4 to 256. (α: ratio between no. synthetic data and no. original training data. $\alpha \in [4, 16, 64, 256]$)

than the others. On using different multi-sensor features, when more synthetic feature vectors are generated for training, different classifiers can obtain consistent performance improvements, except for SVM and "CNN C4-4 H4" on using force_xyz. This demonstrates the usefulness and effectiveness of using GAN for data augmentation.

The number of synthetic data also influences the performance of the classifier. Ideally, if GAN learns to generate new data strictly following the distribution of the original data, and if the original data can be well separated into different classes, then more synthetic data should result in better classification performance. However, the



(a) Sensor channel vib_xyz (b) Sensor channel force_xyz (c) Sensor channel vib_xyz & force_xyz & AE

FIGURE 7. Multi-sensor approach with and without GAN-based synthetic training data. Horizontal axis: different classifiers. Vertical axis: classification accuracy. Blue bar: classification accuracy without GAN. Orange bars: classification accuracy aided by GAN. Color darkness: colors varying from lighter to darker represent α varying from 4 to 256. (α: ratio between no. synthetic data and no. original training data. $\alpha \in [4, 16, 64, 256]$)

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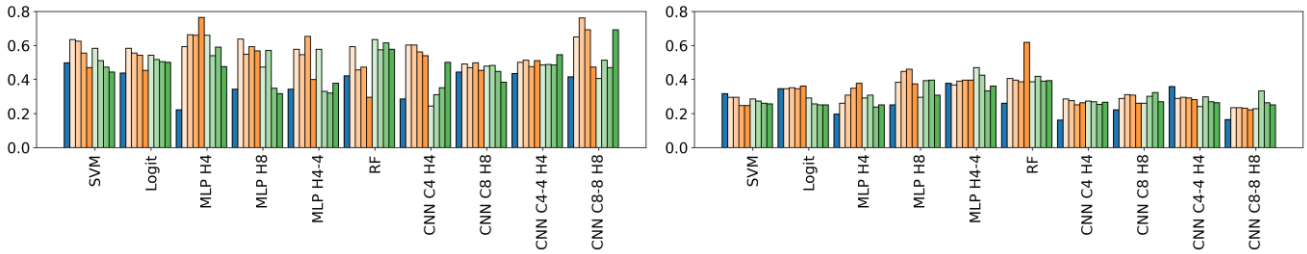


(a) Sensor channel vib_xyz (b) Sensor channel force_xyz

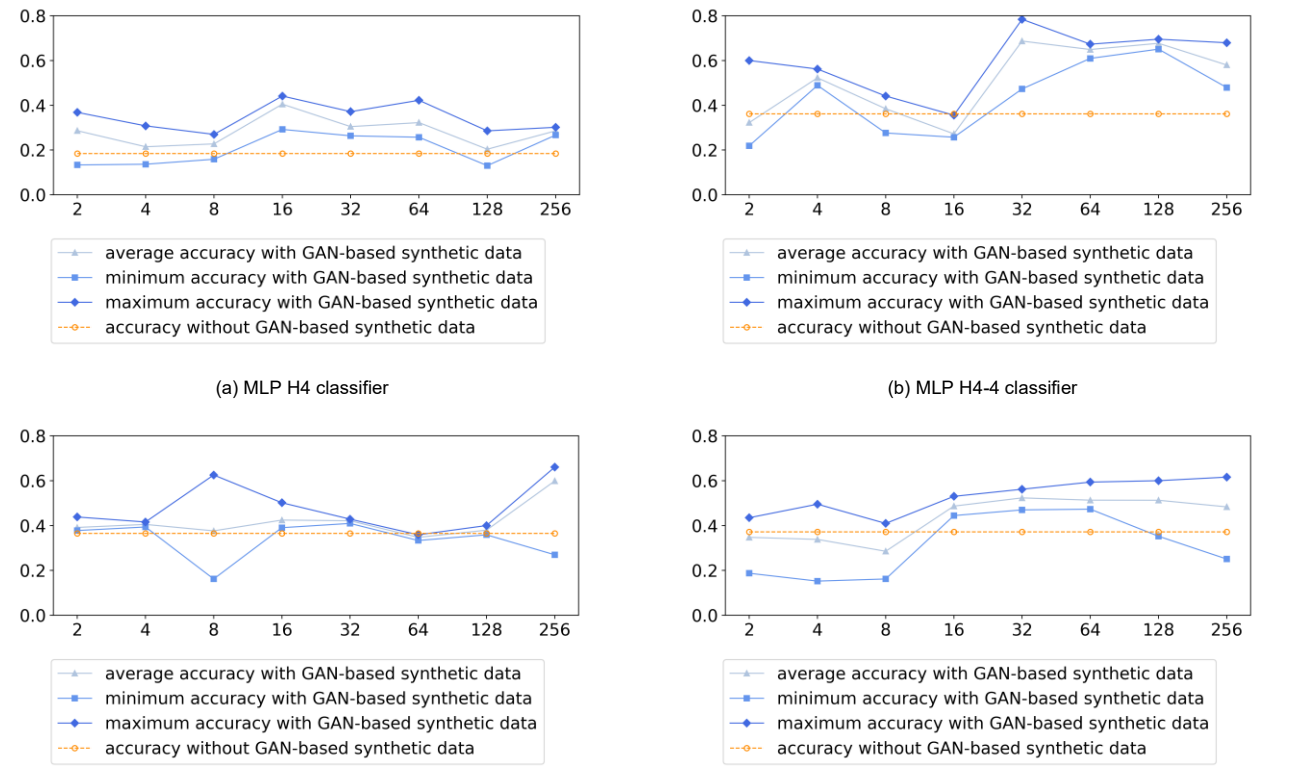
FIGURE 8. Effectiveness of GAN with and without early stopping. Horizontal axis: different classifiers. Vertical axis: classification accuracy. Blue bar: classification accuracy without GAN. Orange bars: classification accuracy aided by GAN (trained with early stopping). Green bars: classification accuracy aided by GAN (trained until equilibrium). Color darkness: colors varying from lighter to darker represent α varying from 4 to 256. (α: ratio between no. synthetic data and no. original training data. $\alpha \in [4, 16, 64, 256]$)

aforementioned two assumptions may not be satisfied in real situations. Therefore, GAN may not be well trained (this may depend on the structure of the generator and the discriminator), and different classes may have overlaps. In this situation, more synthetic data may lead to worse classification boundaries.

4) GAN INFLUENCE FACTOR: EARLY STOPPING

In order to demonstrate the necessity of the early stopping strategy, the effectiveness of the GANs trained with and without early stopping is compared. The feature vectors are extracted from channel vib_xyz and channel force_xyz, on using which the GAN can reach equilibrium (since early stopping is a must if equilibrium cannot be reached). The experimental results are shown in Fig. 8.

As can be seen from Fig. 8, in general, the GANs trained with early stopping outperform those without early stopping (i.e., trained until equilibrium) on using different classifiers, except for a few cases, which may result from the feature characteristics of different sensor types. Therefore, it is believed that early stopping can be a suitable strategy for training effective GANs, since this strategy helps introduce more variations to the synthetic data, which improves the robustness of the classifier.



(a) MLP H4 classifier (b) MLP H4-4 classifier

(c) CNN C4 H4 classifier (d) CNN C4-4 H4 classifier

FIGURE 9. Quality variation of synthetic data with respect to the generation ratio. Horizontal axis: ratio between no. synthetic data and no. original data. Vertical axis: classification accuracy.

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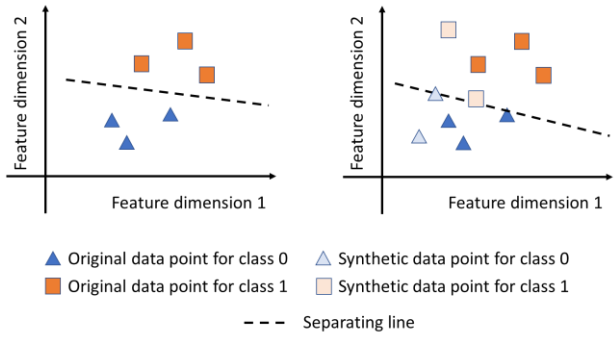

FIGURE 10. Classification performance of a linear classifier trained with and without synthetic data. Left: training using only original data. Right: training using both original and synthetic data.

C. LIMITATION OF GAN

The input to the generator of GAN is randomly sampled from a probability distribution (in this paper, the standard Gaussian distribution). Because of this randomness, the quality of synthetic data may vary. This is an intrinsic limitation of GAN [35]. An illustration of this quality variation is shown in Fig. 9, where the quality of synthetic data is evaluated by evaluating the classification accuracy when a classifier is trained using these synthetic data. The feature vector is extracted from the combination of sensor channels vib_xyz, force_xyz, and AE. For each data generation ratio (i.e., the ratio between the number of synthetic data and the number of original data), synthetic data are randomly generated for 20 times. The average accuracy, minimum accuracy, and maximum accuracy achieved by using these 20 sets of randomly generated data are calculated and plotted in Fig. 9. It can be seen from the figure that, among different sets of randomly generated synthetic data, some sets of data can benefit the training of the classifier and improve the classification accuracy, while some sets of data may harm the training of the classifier and lower the classification accuracy. This indicates the instability of the data generation process.

The structure of GAN is special. GAN consists of a generator and a discriminator competing during training. Even if the training eventually reaches equilibrium, namely both the generator and the discriminator cannot be further improved, it is still possible that both are not well trained. For instance, if the discriminator is too weak to differentiate synthetic data from real data, even in equilibrium, the generator may still produce low-quality synthetic data, as these synthetic data are already "real" enough to deceive the discriminator. Therefore, equilibrium only indicates losing the capability of being further improved, but cannot guarantee having been well trained.

The quality of the original data will also affect the effectiveness of GAN. If the number of usable data for training the GAN is insufficient, GAN may not well learn the characteristics of the data. For classification tasks, if different classes are not well separated, generating more synthetic data for each class may even degrade the classification performance. An example of such degradation in binary classification tasks caused by synthetic data is shown in Fig. 10, where synthetic data may result in overlaps between different classes. Fig. 10 illustrates an example of using a linear classifier to classify 2D features into two classes, by drawing a separating line in the 2D space. As can be seen from the figure, it is possible that the generated synthetic data lead to the failure of separating the data into correct classes.

In practice, the quality of GAN can be evaluated by introducing an additional validation dataset. The high-quality synthetic data can be kept based on the evaluation results. However, this may still be challenging in real industrial environments, where usable data are usually insufficient. Besides, improving the class separability (i.e., making feature vectors belonging to the same class closer to each other, and feature vectors belonging to different classes farther from each other) by extracting more representative feature vectors may improve the robustness of GAN, because this reduces the within-class variations so that GAN can better learn the common characteristics of a specific class of data.

IV. CONCLUSION

This paper compares single-sensor features and multi-sensor features for the prediction of the CNC milling tool wear. Different sensor types, different sensor channels, and different classifiers have been investigated. A simple multi-sensor fusion method is proposed to combine sensor data coming from different sensor channels. This method computes the frequency vector for each sensor channel and then concatenates different frequency vectors to form a higher-dimensional feature vector. Performing feature extraction in the frequency domain will retain most information about the raw sensor data. Dimensionality reduction can be achieved by performing linear filtering on the raw FFT coefficients. It is found that different classifiers behave differently on using different features. The experimental results on a public dataset show that, in general, RF, MLP and CNN outperform SVM and logit when using single-sensor features, while SVM and logit can outperform RF, MLP and CNN when using multi-sensor features. Among different types of sensors, the vibration sensor tends to give the best performance.

Besides the comparison on single-sensor and multi-sensor features, this paper also proposes to utilize GAN for generating more synthetic feature vectors to better train the classifiers. During the training of GAN, an early stopping strategy is implemented, which prevents the training of GAN from being divergent or simply memorizing the original data. The experimental results on a public dataset show that the proposed GAN has the potential of significantly improving the classification performance of different classifiers on

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using single-sensor features and multi-sensor features. Nonetheless, generating a synthetic feature vector requires a random input vector, therefore, there exist quality variations in the synthetic data due to the input randomness. In some cases, the synthetic feature vectors may degrade the performance of the classifier. In addition, the number of synthetic data does not always follow the credo "the more the better". The performance highly depends on the choice of features and classification models. Nevertheless, in practice the quality of the synthetic data can be evaluated by an additional validation dataset, in order to only keep those high-quality synthetic data. Besides, improving the class separability (i.e., making the data belonging to different tool wear states easily separable by some classification models) by extracting highly representative feature vectors from highly effective sensor data may also improve the robustness of GAN and the quality of the synthetic data, and thereby improve the tool wear classification performance.

The proposed GAN can also be used as an add-on to an existing tool wear prediction system without causing interference to the operation of the existing system. Only the extracted feature vectors are required to train the GAN to generate more synthetic feature vectors. Besides, the proposed GAN is also independent of the choice of the tool wear classification model. This provides superior flexibility.

ACKNOWLEDGMENT

We would like to acknowledge the helpful discussions with Florian Clemens from Institute for Industrial Management at RWTH Aachen University, Max Wittstamm from Center Connected Industry at RWTH Aachen Campus (CCI), and Weimin Wang from FLAIR.

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This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2023.3311269

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